Geongyu Lee 이건규 · イ・ゴンギュ

AI Researcher — Computational Pathology × Multi-Omics

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Five years building deep learning for computational pathology and drug discovery. First-author work on H&E-based breast-cancer recurrence (Scientific Reports 2025), a contribution to large-scale pathology foundation models (G2L, AAAI 2026 oral), and KFDA-track model validation. Coming from information security into medical AI, I take a robustness-first approach — calibration, uncertainty, and out-of-distribution behavior matter as much as accuracy once a model reaches the clinic.

SELECTED PUBLICATIONS

1. **Recurrence risk prediction in early-stage breast cancer from H&E whole-slide images.** Scientific Reports, 2025. First author · arXiv:2406.06650
2. **G2L: From Giga-scale to Cancer-specific Pathology Foundation Models via Knowledge Distillation.** AAAI 2026 Workshop (W3PHIAI), Oral. arXiv:2510.11176
3. **MurSS: Multi-Resolution Selective Segmentation for breast cancer.** Bioengineering, 2024. MDPI
4. **AI-driven digital pathology in urological cancers (Review).** Prostate International, 2025. Co-first author
5. **Multi-section WSI analysis for biochemical recurrence in prostate cancer.** Preprint, 2026. arXiv:2603.20273
6. **Supervised contrastive embedding for medical image segmentation.** IEEE Access, 2021.
7. **Glomerular segmentation, patch-to-slide level (KPIs 2024 Challenge, MICCAI).** 2nd place, 2024. arXiv:2502.07288

EXPERIENCE

OMIXAI (fmr. RadiSen) — Feb 2025 – Present

AI Researcher · Seoul

- Lead multi-omics foundation-model R&D integrating proteomics with H&E pathology for drug-response prediction; co-authored the G2L pathology foundation model.
- Built a proteomics drug-response model (Leave-Drug-Out Pearson ≥ 0.65) and a canine veterinary-oncology drug-recommendation algorithm (top-k ≥ 70%).
- Advanced ADMET prediction and self-supervised proteomic representation learning with LoRA / PEFT.
- Co-led the Virtual Cell Challenge — predicting CRISPR-knockdown response in pluripotent stem cells.

Deep Bio — Mar 2021 – Jan 2025

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- First-authored the multi-center breast-cancer recurrence study (Scientific Reports 2025); built a scalable WSI pipeline handling 500+ slides.
- Developed lymph-node metastasis detection and the KPI 2024 glomeruli segmentation model (2nd place, MICCAI).
- Led prostate metastasis & recurrence-risk projects; presented at AACR (2022, 2023) and USCAP (2022).
- Contributed to KFDA regulatory documentation and validation, plus internal server-automation tooling (Docker).

Nuricon — 2021

Intern · Pangyo

- Built a parking-lot fire-detection AI system.

SELECTED PROJECTS

Proteomics drug-response prediction · OMIXAI

Self-supervised proteomic representation with LoRA/PEFT and cell-line IC50 prediction; leave-drug-out validation (Pearson ≥ 0.65).

Veterinary oncology CDSS & Virtual Cell · OMIXAI

Canine drug-recommendation (top-k ≥ 70%) and CRISPR-knockdown response prediction in pluripotent stem cells.

Oncotype DX recurrence prediction · Deep Bio
Breast-cancer prognosis from H&E WSIs only; confidence-aware selection + majority voting into GHI-RS risk groups (high-risk acc 91.2%).

Brain-hemorrhage detection on CT · SK / Ajou Univ. Hospital
2D & 3D hemorrhage segmentation, slice-level classification, domain-generalization validation with class-activation-map auditing.

EDUCATION

M.S. Data Science — Seoul Nat'l University of Science and Technology (SeoulTech) 2019 – 2021
Advisor: Prof. Sangheum Hwang · Thesis: contrastive loss for segmentation under uncertain medical-image labels.

B.S. Information Security — Daejeon University 2012 – 2019
Cross-domain foundation behind a robustness-first approach.

SKILLS

Deep Learning	PyTorch, HuggingFace, PEFT / LoRA, Accelerate, DDP / FSDP
Pathology	OpenSlide, QuPath, CLAM / weakly-supervised MIL, UNI · CONCH · Virchow, WSI tiling
Bio / Omics	scanpy · AnnData, single-cell perturbation, proteomic representation, ADMET
MLOps & Languages	Python, R, SQL, Bash, Docker, Slurm, W&B, MLflow, FastAPI, Git

AWARDS & FUNDING

KPI Challenge 2024 — Whole-Slide Track
2nd place · Glomeruli segmentation, held at MICCAI 2024

Pan-Sarcoma Proteogenomic Profiling for Precision Oncology
Korea–US–Japan · Kyung Hee Univ. Medical Center · 2025–2028 · KRW 400M · Participating Researcher

Virtual-Cell CDSS for Veterinary Oncology
IPET / Ministry of Agriculture · 2026–2030 (awarded) · KRW 3B+ · Participating Researcher

General-Purpose AI for Cancer Pathology Diagnosis
Lead: Deep Bio · National R&D · 2021–2025 · KRW 2.375B · Participating Researcher

TALKS & ABSTRACTS

AACR Annual Meeting (2022, 2023) · USCAP Annual Meeting (2022) · KIIE Fall Conference (2020) — breast-cancer recurrence, protein-receptor status from H&E, cross-organ adenocarcinoma survival, and histological grading.